DB Browser for SQLite - C:\Users\you\Documents\CharlesRiver_Work.sqlite

New Database | Open Database | Write Changes | Revert Changes

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Open SQL file(s) | Save SQL file | ▶ Execute all/selected SQL

Ex 10.4.sql

```
144 -- Requirement (4): amount invoiced against orders and variance by year, 2024 to 2026
145 -- Population: invoice lines with a purchase-order line, dated 2024 to 2026; expected: 2024 0.48%, 2025 0.47%, 2026 0.51%
146 SELECT SUBSTR(pi.InvoiceDate, 1, 4) AS InvoiceYear,
147     COUNT(*) AS InvoiceLines,
148     ROUND(SUM(pil.Quantity * pil.UnitCost), 2) AS AmountInvoiced,
149     ROUND(SUM(pil.LineTotal), 2) AS LineTotals,
150     ROUND(SUM(pil.Quantity * (pil.UnitCost - pol.UnitCost)), 2)
151         AS PriceVariance,
152     ROUND(100.0 * SUM(pil.Quantity * (pil.UnitCost - pol.UnitCost))
153         / SUM(pil.Quantity * pil.UnitCost), 2) AS VariancePct
154 FROM PurchaseInvoiceLine AS pil
155     INNER JOIN PurchaseInvoice AS pi
156         ON pi.PurchaseInvoiceID = pil.PurchaseInvoiceID
157     INNER JOIN PurchaseOrderLine AS pol
158         ON pol.POLineID = pil.POLineID
159 WHERE pi.InvoiceDate BETWEEN '2024-01-01' AND '2026-12-31'
160 GROUP BY InvoiceYear
161 ORDER BY InvoiceYear;
```

	InvoiceYear	InvoiceLines	AmountInvoiced	LineTotals	PriceVariance	VariancePct
1	2024	6898	13491998.13	13491998.93	64631.92	0.48
2	2025	13529	17468202.42	17468203.84	82377.43	0.47
3	2026	19866	21612260.34	21612262.33	109566.2	0.51

Execution finished without errors.
Result: 3 rows returned
At line 146:
SELECT SUBSTR(pi.InvoiceDate, 1, 4) AS InvoiceYear,

Outlined: VariancePct, 0.48%, 0.47% and 0.51%, a near-steady share of what is invoiced against orders while that amount grows 60% from 2024 to 2026. The variance grows with the amount invoiced. AmountInvoiced mixes quantity, order prices and mix, so the comparison cannot say how much of the growth is more buying. LineTotals differs from the amount by the rounding of each line.